

SHEPPARTON, Australia

WMO: 948750

Lat: 36.4289S

Lon: 145.3947E

Elev: 376

StdP: 14.50

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.9	33.1	27.2	21.4	63.7	29.9	24.3	53.9	22.1	53.8	20.2	52.9	1.1	160	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	99.8	66.2	95.2	65.1	91.3	64.7	71.3	84.2	69.4	83.6	67.8	83.8	11.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	104.7	74.1	65.4	95.1	73.6	62.9	87.0	71.5	35.3	83.8	33.7	83.2	32.3	83.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.9	17.9	DB	27.1	106.3	1.7	2.9	25.9	108.4	24.9	110.1	24.0	111.7	22.8	113.8	(4)
(5)			WB	27.0	74.2	1.7	1.8	25.7	75.5	24.7	76.5	23.7	77.5	22.5	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	73.9	72.7	67.7	59.8	52.6	47.9	47.0	48.3	52.7	58.2	64.9	69.3
	DBStd	11.07	7.16	6.33	6.40	5.53	4.53	3.96	3.72	3.90	4.95	6.08	6.89	6.99
	HDD50	326	0	0	0	2	24	87	107	80	23	3	0	0
	HDD65	2917	8	9	42	173	386	514	558	517	369	226	83	32
	CDD50	3799	742	636	550	295	104	22	14	28	105	259	446	598
	CDD65	914	285	225	127	16	0	0	0	0	1	16	80	164
	CDH74	12240	3845	2675	1486	231	2	0	0	0	38	380	1244	2339
	CDH80	5802	2099	1299	595	35	0	0	0	0	7	122	520	1125
Wind	WSAvg	8.3	10.2	9.4	8.6	6.9	6.5	6.3	7.0	7.6	8.2	8.8	9.5	10.2
Precipitation	PrecAvg	19.8	1.5	1.1	1.2	1.6	1.9	1.7	2.0	2.0	2.0	1.9	1.5	1.5
	PrecMax	35.2	5.0	4.8	4.3	6.0	4.9	5.6	4.8	4.4	5.1	7.1	4.1	5.2
	PrecMin	8.5	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.2	0.4	0.2	0.0	0.0
	PrecStd	6.1	1.3	1.3	1.1	1.3	1.3	1.1	1.0	1.0	1.1	1.6	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.6	103.3	97.3	84.3	72.5	64.1	62.2	67.7	78.7	89.9	96.3	102.6
		MCWB	66.8	68.3	66.1	61.4	57.5	54.4	53.4	56.0	58.0	60.7	64.3	64.9
	2%	DB	101.0	96.2	90.7	79.6	68.5	60.7	58.6	63.2	72.4	83.1	90.8	96.3
		MCWB	66.5	66.4	64.9	60.6	56.7	52.9	51.8	53.5	57.6	59.6	62.8	63.9
	5%	DB	95.9	91.8	86.6	76.4	65.4	58.5	56.7	60.2	68.0	77.7	86.0	91.0
		MCWB	65.9	66.0	64.0	59.6	55.7	52.1	51.0	52.1	56.3	58.6	62.0	63.4
	10%	DB	91.2	88.1	82.4	72.9	62.6	56.4	54.8	57.5	64.4	73.1	81.2	86.2
		MCWB	65.2	64.9	62.9	58.5	54.7	51.2	49.8	50.8	54.7	57.2	61.1	62.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	74.5	70.7	65.5	61.7	57.2	55.8	57.4	62.5	65.5	69.7	71.1
		MCDB	83.1	88.6	81.1	76.1	65.4	60.4	58.4	64.4	70.6	78.3	83.0	80.0
	2%	WB	71.2	71.5	68.3	63.1	58.8	54.8	53.3	55.1	59.9	62.8	66.7	68.4
		MCDB	86.8	83.3	81.9	72.4	64.3	58.5	56.5	60.9	67.9	73.9	79.9	81.4
	5%	WB	69.2	69.4	66.3	61.4	57.0	53.4	51.8	53.0	57.6	60.9	64.8	66.4
		MCDB	86.8	82.8	80.6	71.5	63.1	56.8	55.3	58.5	65.6	71.4	77.6	82.2
	10%	WB	67.3	67.5	64.7	59.8	55.5	51.9	50.5	51.3	55.4	58.9	63.0	64.6
		MCDB	86.6	82.4	79.0	70.7	61.3	55.4	53.8	56.6	63.3	71.0	76.3	82.0
Mean Daily Temperature Range	5% DB	MDBR	28.1	26.3	24.9	23.6	20.1	17.3	16.1	18.7	21.9	25.0	26.4	27.3
		MCDBR	35.5	32.1	30.7	29.0	24.1	20.8	18.5	23.4	28.1	33.6	34.5	35.0
		MCWBR	10.1	9.8	10.3	12.7	13.6	13.7	12.4	14.6	15.0	14.3	11.7	10.8
	5% WB	MCDBR	27.9	26.0	24.5	23.0	19.4	16.1	15.5	20.2	23.9	26.1	25.3	26.9
		MCWBR	10.1	10.2	10.1	11.2	11.9	11.5	10.9	13.3	14.2	13.4	11.0	10.8
Clear-Sky Solar Irradiance	taub	0.346	0.345	0.322	0.313	0.301	0.293	0.290	0.300	0.328	0.330	0.344	0.337	
	taud	2.474	2.492	2.552	2.546	2.555	2.576	2.583	2.532	2.442	2.460	2.459	2.494	
	Ebn at Noon	313	306	302	285	271	265	273	286	295	308	312	318	
	Edh at Noon	37	35	30	27	23	21	22	27	33	36	37	36	
All-Sky Solar Radiation	RadAvg	2453	2166	1774	1264	862	659	712	1009	1432	1892	2201	2449	
	RadStd	128	116	80	59	56	76	39	77	123	151	110	108	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209
(47) Regional (31 neighbors)		+0.90	N/A	N/A	+1.71	N/A	N/A	-36	-212	+313	+128

Nomenclature: See separate page